RICHFIELD GRADUATE INSTITUTE OF TECHNOLOGY

Faculty of Information Technology

Bachelor of Science in Information Technology (BSc IT) | IT Project (Year 3), Semester 2 2026

HireTrack

Applicant Tracking & Resource Allocation System

Phase 5: Implementation (Sections 5.1 to 5.5)

Item	Detail
Project title	HireTrack: Applicant Tracking & Resource Allocation System
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Qualification	BSc IT, IT Project (Year 3)
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Date	11 November 2026

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5.1 Introduction

5.1.1 Purpose of the Implementation Phase

The Implementation Phase turns the System Design (Phase 4, 30 October 2026) into working, tested and installable software. It answers three questions: does the system do what the requirements say, does it do it reliably and fast enough, and can someone else install and run it from the submitted files alone.

This document covers the five parts the Faculty guidelines require for this phase: coding (5.2), testing (5.3), system testing with test cases and an evaluation of the results (5.4), and installation (5.5).

5.1.2 Scope of this release

The Design phase narrowed the first release to the candidate-job matching engine and listed user accounts, the applicant pipeline and the recruiter dashboard as future scope. The Proposal (28 August 2026), however, described HireTrack as a full applicant tracking and resource allocation system. Because the matching engine was finished early, the group brought the Proposal's remaining objectives into this release. The matching engine was kept exactly as designed and became one module of the larger system.

Area	Status in Design (Phase 4)	Status in this release
Candidate and job management, four-part scoring, ranking, CSV export, backups	In scope	Built as designed; job location and open/closed status added
User accounts and roles (JWT + bcrypt)	Future scope	Built: Administrator, Recruiter (HR), Resource Manager
Applicant pipeline with audit history	Future scope	Built: Applied, Screened, Interview, Offer, Hired, Rejected
Recruiter dashboard	Future scope	Built: key counts and five charts
Uploading CV files	Future scope	Built: PDF, Word and text CVs pre-fill the form
Bulk import	Not planned	Built: CSV and Excel with a per-row error report
Resource allocation (staff, projects, utilisation)	Proposal title only	Built: allocations by percentage and date with no over-booking
Multi-user access over a network	Future scope	Still future scope (runs on 127.0.0.1)

Table 5.1: Scope of the release compared with the Design phase

The requirements were extended to match. FR1 to FR13 and NFR1 to NFR8 from the Analysis phase still apply. FR14 to FR26 and NFR9 to NFR10 were added for the new modules and are listed in Section 5.4.1.

5.1.3 Technology stack and reasons

The stack is the one chosen in the Planning and Design phases. No paid service or cloud platform is used.

Layer	Technology	Why it was chosen
Frontend	React 18, Vite 5, Tailwind CSS 3	Component model suits many similar screens; Vite builds fast; Tailwind keeps one consistent visual style without a heavy component library.
Backend	Python 3.11, FastAPI, Pydantic 2, Uvicorn	The matching engine and the model library are Python. FastAPI validates every request from type hints and generates API documentation at /docs.
Data access	SQLAlchemy 2 ORM	Bound parameters only (no SQL injection), and the same models define the tables and their constraints.
Database	SQLite	One file, no server to install, supports foreign keys and CHECK constraints, easy to back up.
Matching	sentence-transformers all-MiniLM-L6-v2	Small (about 90 MB), runs on an ordinary CPU, works offline, free. Finds candidates who describe the work in different words.
Security	bcrypt (passlib), PyJWT	Industry-standard password hashing and signed, expiring sessions, as planned in the Proposal.
Import	openpyxl, pypdf	Read Excel files and PDF CVs without Microsoft Office.
Testing	pytest, pytest-cov, Vitest, React Testing Library, Playwright	One tool per level: unit and API (pytest), screens (Vitest), full journeys in a real browser (Playwright).

Table 5.2: Technology stack

The system follows the linear life cycle with iterative build cycles chosen in Planning: each module was built, unit tested and reviewed before the next one started, and the full suites were re-run after every change.

5.2 Coding

5.2.1 System architecture

HireTrack is a three-tier client-server system that runs on one computer (Figure 5.1). The browser shows a React single-page application. It talks to a FastAPI server over HTTP using JSON. The server keeps everything in a SQLite file. In normal use the server also serves the built screens, so the whole system uses one port.

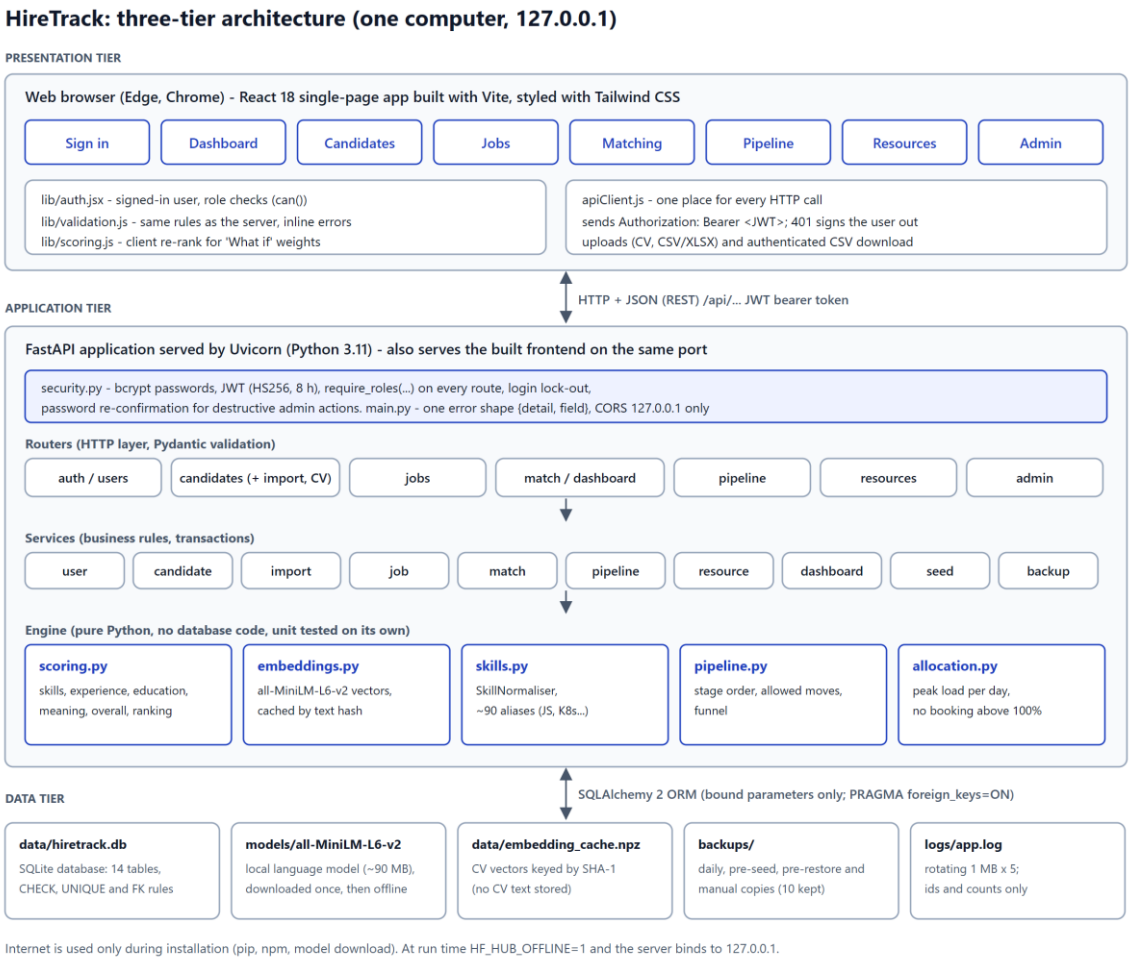


Figure 5.1: Three-tier architecture of HireTrack

Inside the server the code is layered. Routers handle HTTP and validation, services hold the business rules and the database transactions, and the engine modules hold pure rules with no database code. This means the rules that matter most (scoring, stage changes, allocation limits) can be tested on their own in milliseconds.

5.2.2 Module breakdown

Module	Main files	Responsibility	Requirements
M1 Accounts and security	security.py, routers/auth.py, services/user_service.py, pages/Login.jsx, lib/auth.jsx	Sign-in, sessions, role checks, user administration, password change and lock-out	FR14, FR15, NFR9

Module	Main files	Responsibility	Requirements
M2 Candidate management	services/candidate_service.py, services/import_service.py, pages/Candidates.jsx, ImportPanel.jsx	Candidate CRUD and search; CV text extraction; CSV/Excel import	FR1, FR2, FR12, FR16, FR17
M3 Job specifications	services/job_service.py, pages/Jobs.jsx	Job CRUD with skills, location, status and weights	FR3, FR4
M4 Matching engine	engine/scoring.py, engine/embeddings.py, engine/skills.py, services/match_service.py, pages/Results.jsx	Four scores, overall score, ranking, explanations, what-if re-ranking, CSV export	FR5 to FR10, FR13, NFR1, NFR7
M5 Applicant pipeline	engine/pipeline.py, services/pipeline_service.py, pages/Pipeline.jsx	Applications, stage rules, audit history, shortlist from results, hire into staff pool	FR18 to FR21
M6 Resource allocation	engine/allocation.py, services/resource_service.py, pages/Resources.jsx	Staff, projects, allocations, no booking above 100%, utilisation and bench	FR22 to FR24
M7 Dashboard	services/dashboard_service.py, pages/Dashboard.jsx, components/Charts.jsx	Key counts, funnel, applicants per job, utilisation, hires per month, activity	FR25
M8 Administration	services/seed_service.py, services/backup_service.py, routers/admin.py, pages/Admin.jsx, seed/	Seed datasets, backups, integrity-checked restore	FR11, FR26

Table 5.3: Modules, files and the requirements each one covers

5.2.3 Database schema

The database extends the Phase 4 design (Figure 5.2). The seven tables from the design are unchanged apart from two columns on the job specification (location and status) and two optional contact columns on the candidate. Seven tables were added for accounts, the pipeline and resource allocation. The design is in Third Normal Form: each non-key column depends only on its table's key, lists are stored in link tables, and lookups (education level, skill) are separate tables.

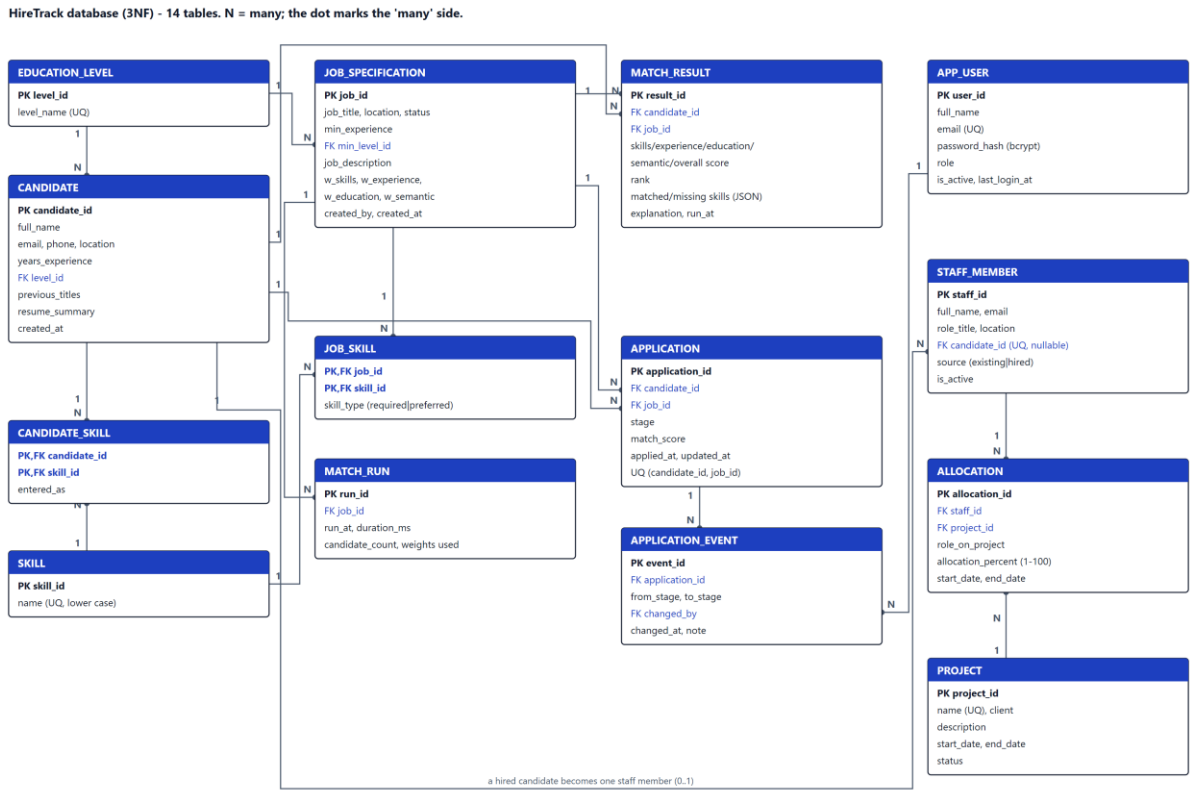


Figure 5.2: Entity-relationship diagram (14 tables)

Table	Purpose	Key rules (enforced by the database)	Rows in seed
app_user	Sign-in accounts	email UNIQUE and lower case; role IN (admin, recruiter, resource_manager)	3
candidate	Talent pool	name 2-100 characters; years 0-50; summary 20-3000 characters	600
job_specification	Roles to fill	location 2-80 characters; status IN (open, closed); weights 0-100 and total = 100	33
application	Candidate applying for a job	stage IN (6 stages); UNIQUE (candidate_id, job_id); cascades with candidate and job	847
application_event	Audit history of stage changes	one row per change; changed_by kept as SET NULL if a user is removed	1906
staff_member	People who can be allocated	candidate_id UNIQUE (a hire becomes one staff member); source IN (existing, hired)	73
project	Client projects	name UNIQUE; end_date >= start_date; status IN (planned, active, on_hold, completed)	12
allocation	Share of a person's time on a project	allocation_percent 1-100; end_date >= start_date; the 100% daily limit is enforced by the service	102

Table 5.4: New and changed tables (the other six are as in the Design phase)

An existing database from the Phase 4 build is upgraded automatically when the server starts: missing columns are added with safe defaults, missing tables are created and the default accounts are added if there are none. Restoring an old backup goes through the same step.

5.2.4 Key code excerpts

The excerpts below are copied from the source files. Each shows a rule the requirements depend on.

a) Skills score (FR5)

Required skills are worth 80 points and preferred skills 20. Skills are compared after normalisation, so "JS" on a CV matches "JavaScript" on a job. The function returns which skills matched and which are missing, which the results screen shows.

backend/app/engine/scoring.py

```
def skills_score(candidate_skills: Sequence[str], required: Sequence[str], preferred:
Sequence[str]) -> SkillMatch:
    """Required skills are worth 80 points, preferred 20.

    With no preferred skills, required skills are worth the full 100.
    Inputs must already be canonical (see SkillNormaliser).
    """
    have = set(candidate_skills)
    matched_req = [s for s in required if s in have]
    missing_req = [s for s in required if s not in have]
    matched_pref = [s for s in preferred if s in have]
    missing_pref = [s for s in preferred if s not in have]

    if not required:
        req_part = 0.0
    elif preferred:
        req_part = len(matched_req) / len(required) * REQUIRED_SHARE
    else:
        req_part = len(matched_req) / len(required) * 100
    pref_part = len(matched_pref) / len(preferred) * PREFERRED_SHARE if preferred else 0.0

    return SkillMatch(
        score=round1(clamp(req_part + pref_part)),
        matched_required=matched_req,
        missing_required=missing_req,
        matched_preferred=matched_pref,
        missing_preferred=missing_pref,
    )
```

b) Overall score (FR9)

The overall score is a weighted average of the four scores. It is calculated in whole tenths with round-half-up, so the browser's instant "what if" re-ranking, which uses the same method in JavaScript, always agrees with the server to the last decimal.

backend/app/engine/scoring.py

```
def overall_score(skills: float, experience: float, education: float, semantic: float, weights:
Weights) -> float:
    """Weighted average of the four components, rounded to 1 decimal.

    Computed in integer tenths so the browser's "what if" re-ranking (which
    uses the same integer method) always agrees with the server exactly.
    """
    tenths = [int(round(c * 10)) for c in (skills, experience, education, semantic)]
    ws = (weights.skills, weights.experience, weights.education, weights.semantic)
    total = sum(t * w for t, w in zip(tenths, ws))  # in (tenths x percent)
    return ((total + 50) // 100) / 10  # round half up back to tenths
```

c) Pipeline stage rules (FR18)

An application moves forward one stage at a time or is rejected. Hired and Rejected are final, except that a rejection can be reopened at the stage it was rejected from. The function raises an error with a reason that the screen shows to the user.

backend/app/engine/pipeline.py

```
def check_move(current: str, target: str, reopen_to: str | None = None) -> None:
    """Raise StageError with a plain-English reason if current -> target is not allowed."""
    if target not in ALL_STAGES:
        raise StageError(f"Unknown stage '{target}'")
    if current == target:
        raise StageError(f"The application is already at {LABELS[current]}")
    if current in FINAL and not (current == REJECTED and target == reopen_to):
        raise StageError(f"{LABELS[current]} is a final stage and cannot be changed")
    if target not in allowed_moves(current, reopen_to):
        nxt = next_stage(current)
        hint = f" The next stage is {LABELS[nxt]}." if nxt else ""
        raise StageError(f"Cannot move from {LABELS[current]} to {LABELS[target]}.{hint}")
```

d) Changing a stage: audit history and hiring (FR19, FR21)

The stage change and its history row are written in the same transaction, so the history can never disagree with the current stage. A move to Hired also creates the staff record.

backend/app/services/pipeline_service.py

```
def change_stage(db: Session, application_id: int, data: StageChangeIn, user: User) -> Application:
    app = get_application(db, application_id)
    try:
        rules.check_move(app.stage, data.stage, _rejected_from(app))
    except rules.StageError as exc:
        raise AppError(str(exc), field="stage", status_code=422) from exc
    if data.stage == "hired" and app.job.status != "open":
        raise AppError("This job is closed, so nobody else can be hired for it", field="stage",
status_code=422)
    now = utcnow()
    app.events.append(
        ApplicationEvent(
            from_stage=app.stage, to_stage=data.stage, changed_by=user.user_id, changed_at=now,
note=data.note or None
        )
    )
    app.stage = data.stage
    app.updated_at = now
    if data.stage == "hired":
        _ensure_staff(db, app)
    db.commit()
    log.info("Application id=%s moved to %s by user id=%s", application_id, data.stage,
user.user_id)
    db.expire_all()
    return get_application(db, application_id)
```

e) Preventing over-allocation (FR23)

A person's load can only change on the day a booking starts or the day after one ends. The check therefore looks only at those days instead of every day in the range, and reports the first day the limit would be broken.

backend/app/engine/allocation.py

```
def peak_load(bookings: Iterable[Booking], start: date, end: date) -> tuple[int, date | None]:
    """Highest total load on any day in [start, end], and the first day it happens.

    Load only changes on a booking's start day or the day after its end, so it
    is enough to check those days (a sweep), not every day in the range.
    """
    inside = [b for b in bookings if overlaps(b.start, b.end, start, end)]
    if not inside:
        return 0, None
    days = {start}
    for b in inside:
        if start <= b.start <= end:
            days.add(b.start)
        after = b.end + timedelta(days=1)
        if start <= after <= end:
            days.add(after)
    best, best_day = 0, None
    for d in sorted(days):
        load = load_on(inside, d)
        if load > best:
            best, best_day = load, d
    return best, best_day
```

f) Role-based access on the server (FR14, NFR9)

Every route declares the roles it allows. The check runs on the server, so hiding a button in the browser is never the only protection.

backend/app/security.py

```
def require_roles(*roles: str):
    """Dependency factory: allow only users whose role is in `roles` (else 403)."""

    def _check(user=Depends(current_user)):
        if user.role not in roles:
            log.warning("Denied user id=%s role=%s (needs %s)", user.user_id, user.role,
"/".join(roles))
            raise Forbidden("Your role does not have access to this action.")
        return user

    return _check
```

backend/app/routers/pipeline.py

```
@router.post("/{application_id}/stage", response_model=ApplicationOut)
```

g) Bulk import validation (FR17)

Each row of an imported file is checked with exactly the same Pydantic model as the Add Candidate form. Bad rows are reported with their row number; good rows are saved together in one transaction.

backend/app/services/import_service.py (import_candidates)

```
for i, row in enumerate(rows, start=2):  # row 1 is the header
    try:
        data = CandidateIn(**_to_input(row))
    except ValidationError as exc:
        field, msg = _friendly(exc.errors()[0])
        errors.append(ImportRowError(row=i, field=field, message=msg))
        continue
    if data.email and (data.email in existing_emails or data.email in seen_in_file):
        skipped += 1
        continue
    if data.email:
        seen_in_file.add(data.email)
    valid.append(data)
```

5.2.5 Coding standards, validation and error handling

- **Naming and layout.** Python follows PEP 8 with type hints; JavaScript uses one component per file. Files are named after what they hold (pipeline_service.py, Pipeline.jsx).
- **Comments.** Every module starts with a docstring explaining its job and its rules. Comments explain why, not what (for example why scores are calculated in tenths).
- **Three layers of validation (NFR6).** The screen checks input as it is typed (lib/validation.js); Pydantic checks every request on the server; database CHECK, UNIQUE and FOREIGN KEY rules are the last line of defence.
- **One error shape.** Every API error is {"detail": "...", "field": "..."}. The screens use the field name to show the message under the right input.
- **Transactions.** Matching, import, seeding and stage changes write everything or nothing; any error rolls back.
- **Parameter passing.** Routers receive validated schema objects and pass them to service functions together with the database session and the signed-in user, so services never read HTTP data directly.
- **Efficiency.** CV embeddings are calculated once and cached in memory and on disk; candidates are loaded with selectinload to avoid one query per row; the candidate table is virtualised so 2,600 rows scroll smoothly.

5.2.6 Security measures

Threat	Control
Stolen passwords	Only bcrypt hashes are stored (12 rounds). Passwords need 8+ characters with a letter and a number.
Guessing passwords	Five failed sign-ins lock that email for 60 seconds. The same message is shown for a wrong email or password.
Using the system without signing in	Every API route except /api/health needs a signed JWT (HS256, 8-hour expiry). The secret is generated on first run and kept in data/.jwt_secret.
Doing more than the role allows	require_roles(...) on every route (36 role and endpoint combinations tested). Screens hide what a role cannot use.
Accidental data loss	Loading a dataset or restoring a backup needs the administrator's password again. A safety backup is taken first; restore runs PRAGMA integrity_check.
SQL injection / XSS	SQLAlchemy bound parameters only; React escapes all text; no dangerouslySetInnerHTML.
Spreadsheet formula injection	CSV export prefixes cells that start with =, +, - or @.
Network exposure	The server binds to 127.0.0.1; CORS allows only the local addresses.
Privacy (POPIA)	Only fictitious data; logs hold ids and counts, never CV text; scoring never reads name, email, age or gender.

Table 5.5: Security controls

5.2.7 User interface

The screens use one neutral palette with a single blue accent, green only for Run Match and red only for destructive actions. Every page uses the same header, table and side-panel patterns. The menu shows only what the signed-in role can use (Figures 5.3 to 5.12).

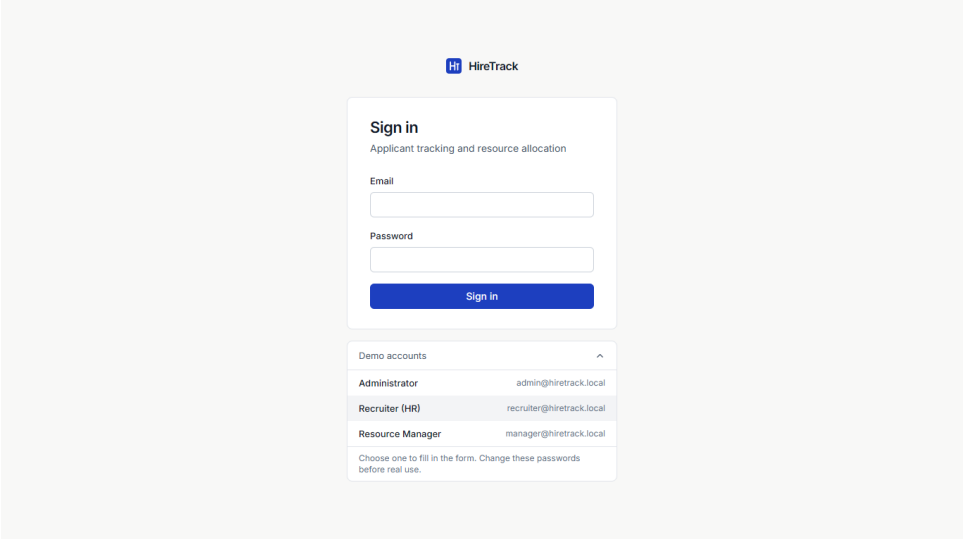


Figure 5.3: Sign-in screen with the demo accounts expanded

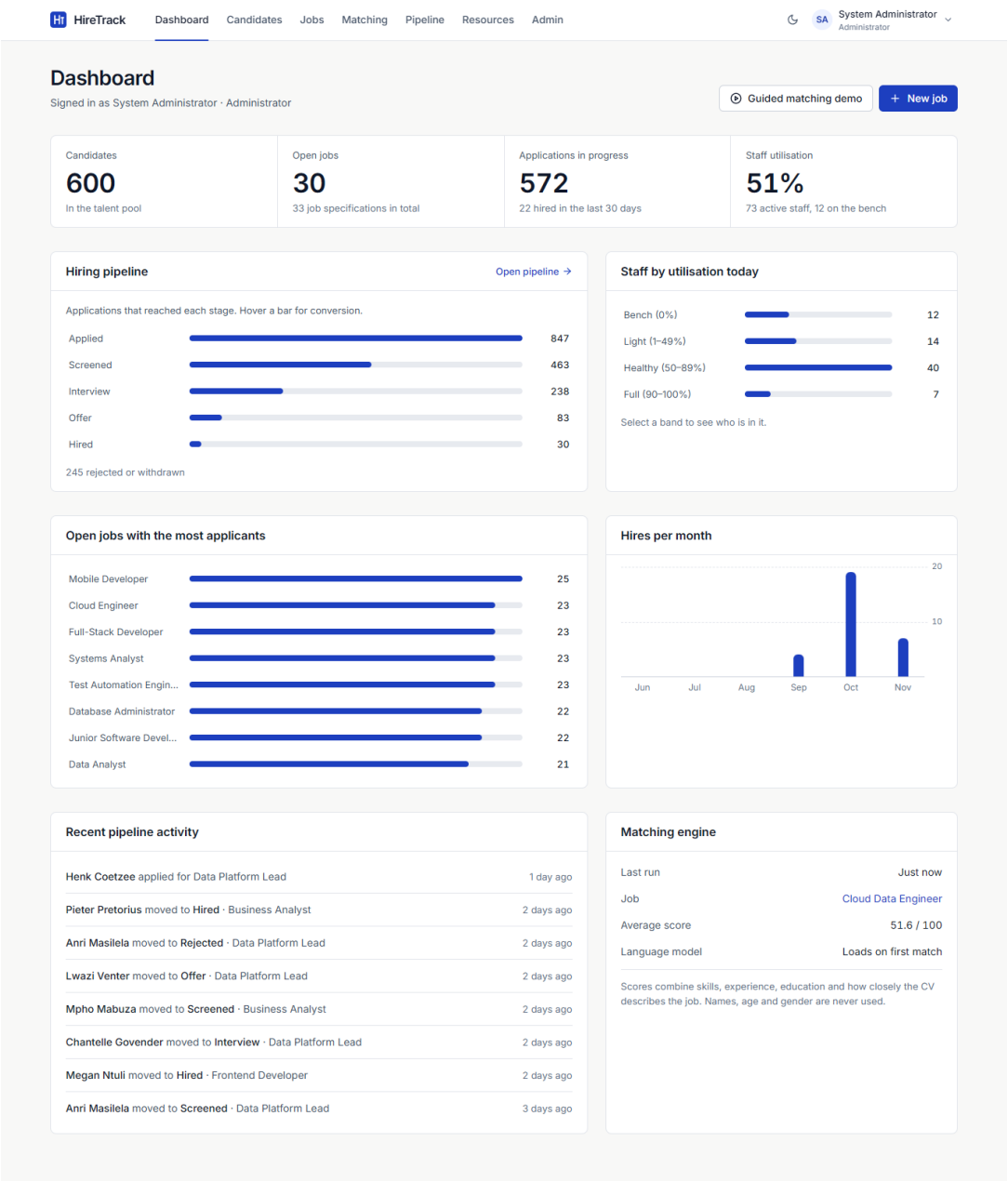


Figure 5.4: Dashboard (Administrator): key counts, pipeline funnel, utilisation, applicants per job, hires per month

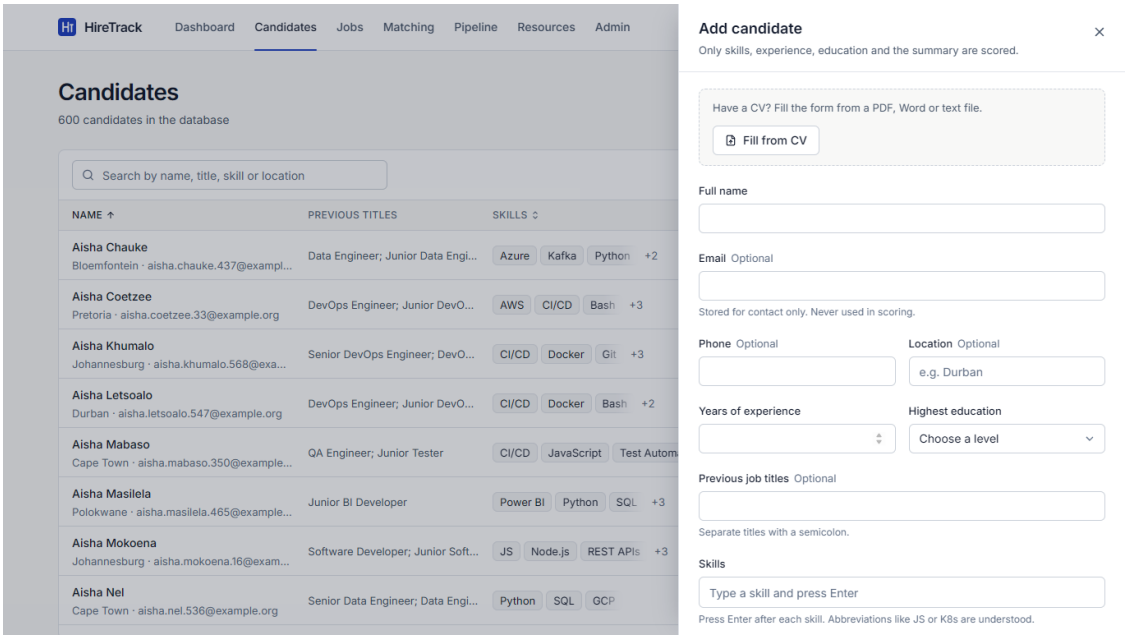


Figure 5.5: Add Candidate panel with "Fill from CV"

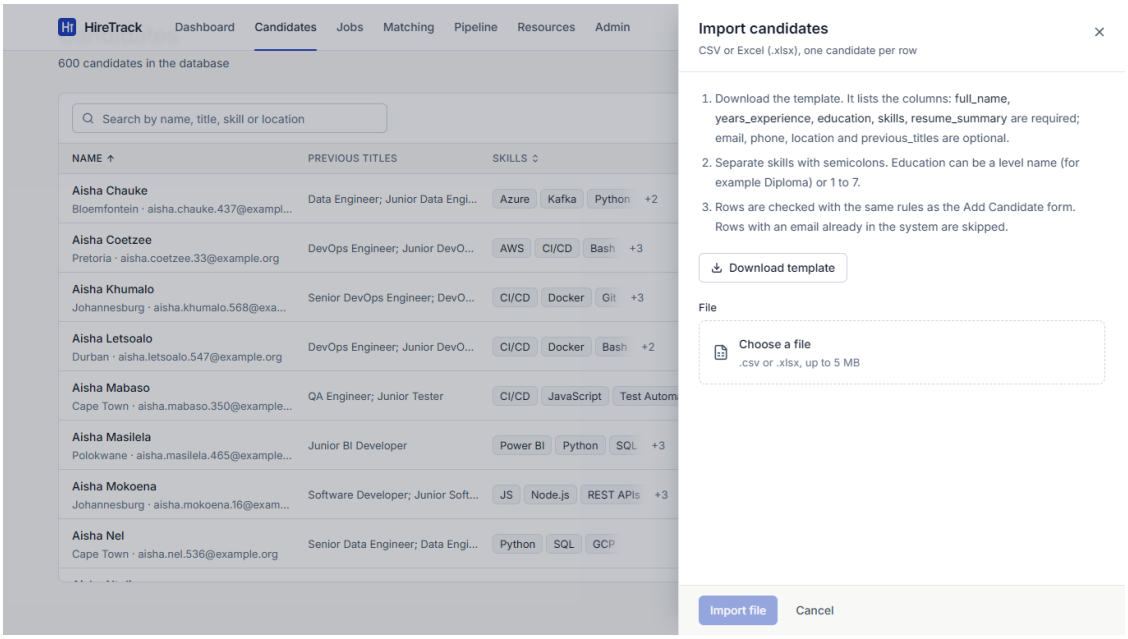


Figure 5.6: Bulk import of candidates from CSV or Excel

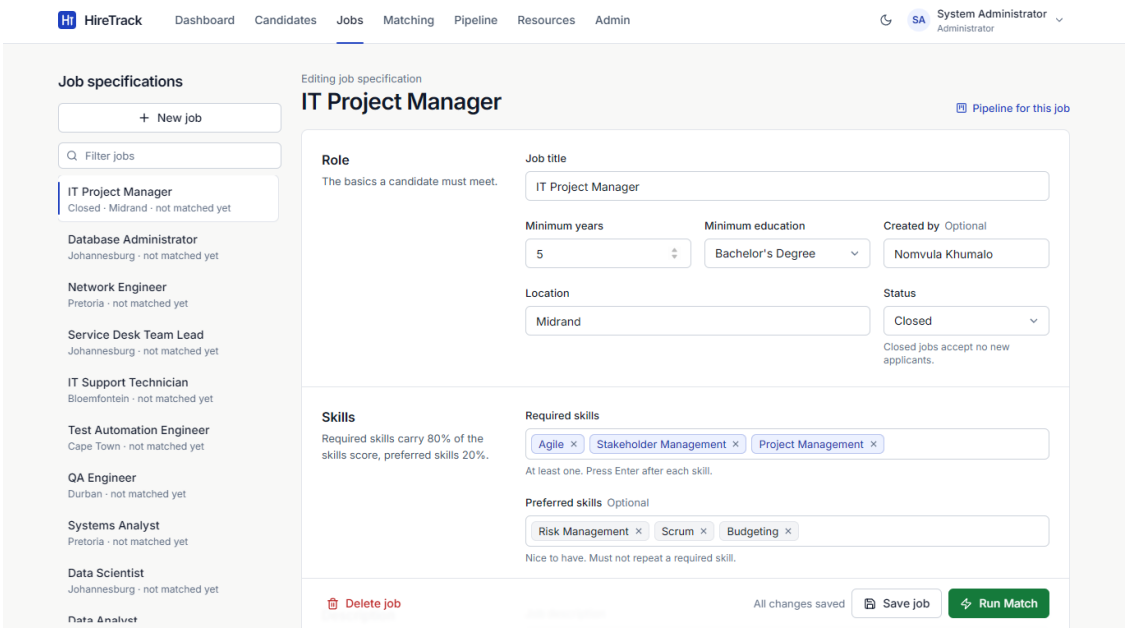


Figure 5.7: Job specification with location, status, skills and weights

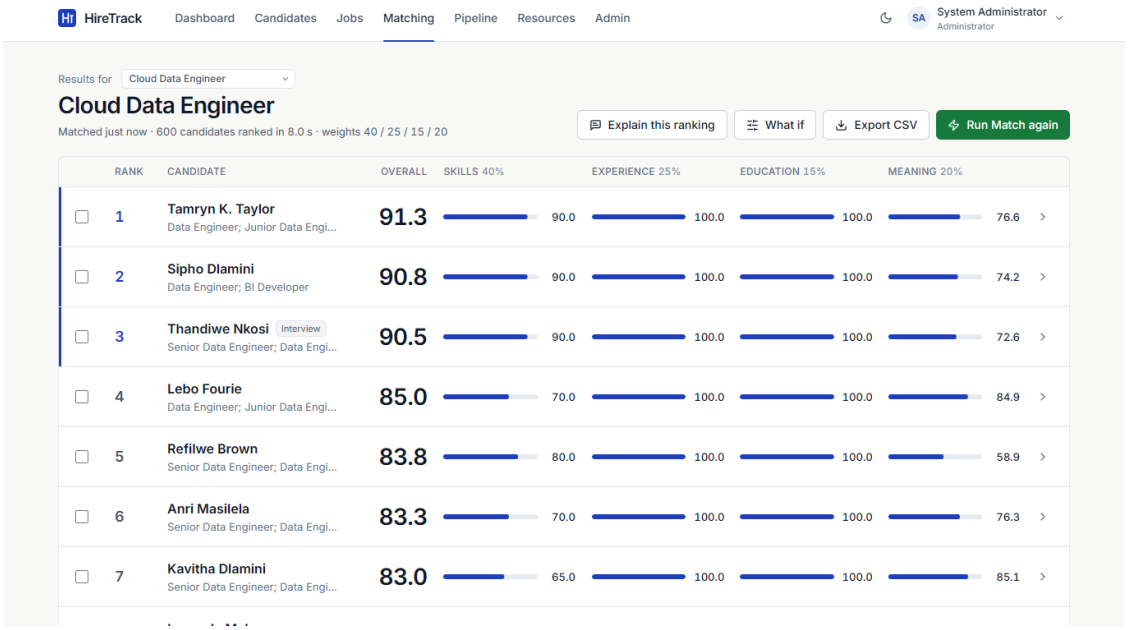


Figure 5.8: Ranked results for Cloud Data Engineer; a pipeline stage badge shows candidates already applied

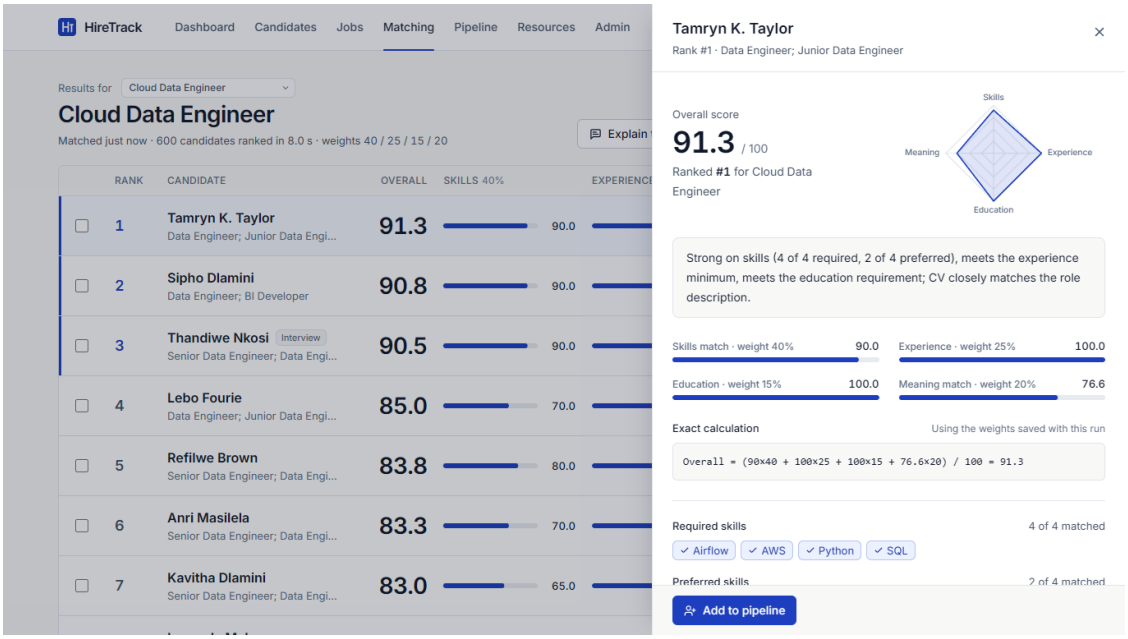


Figure 5.9: Score breakdown with the exact calculation and "Add to pipeline"

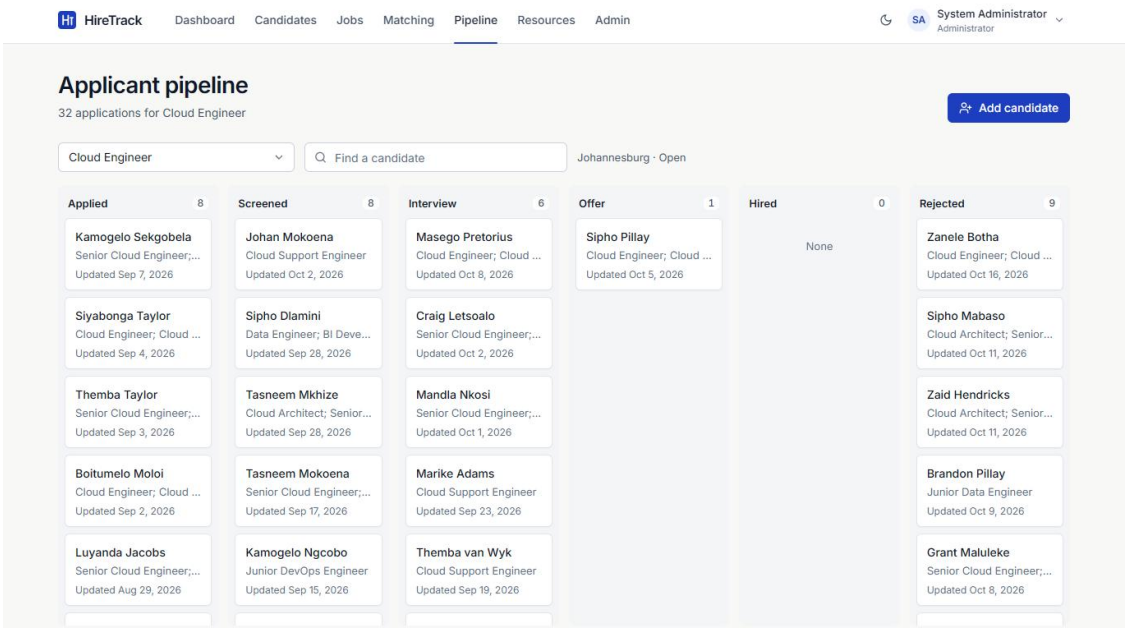


Figure 5.10: Applicant pipeline board for one job

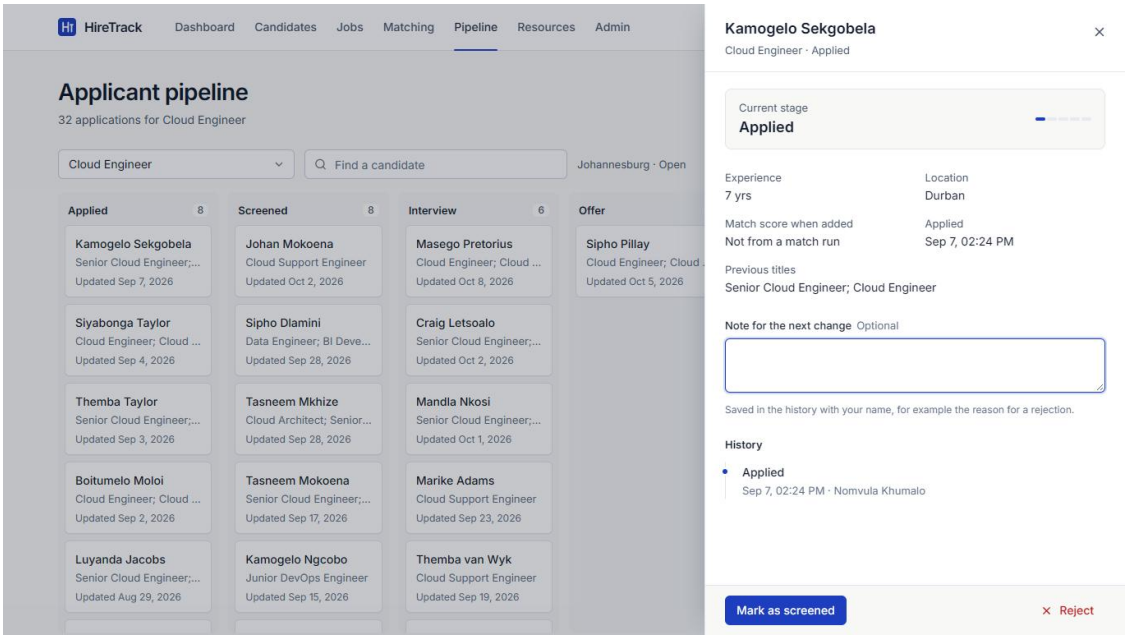


Figure 5.11: Application detail: stage actions, note and audit history

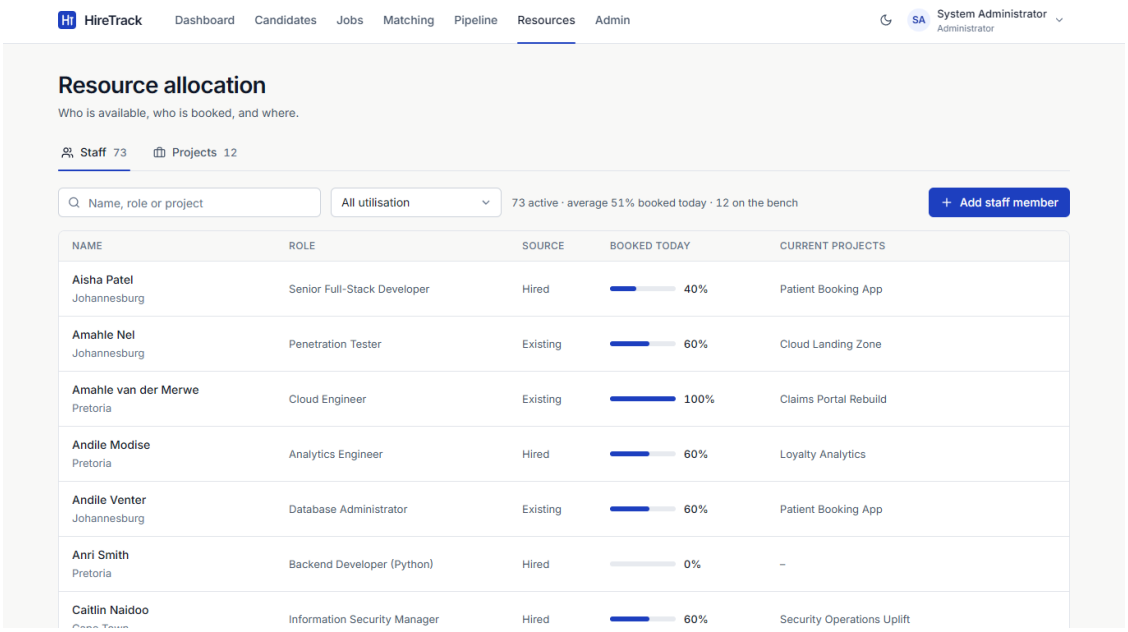


Figure 5.12: Staff utilisation in the Resources screen

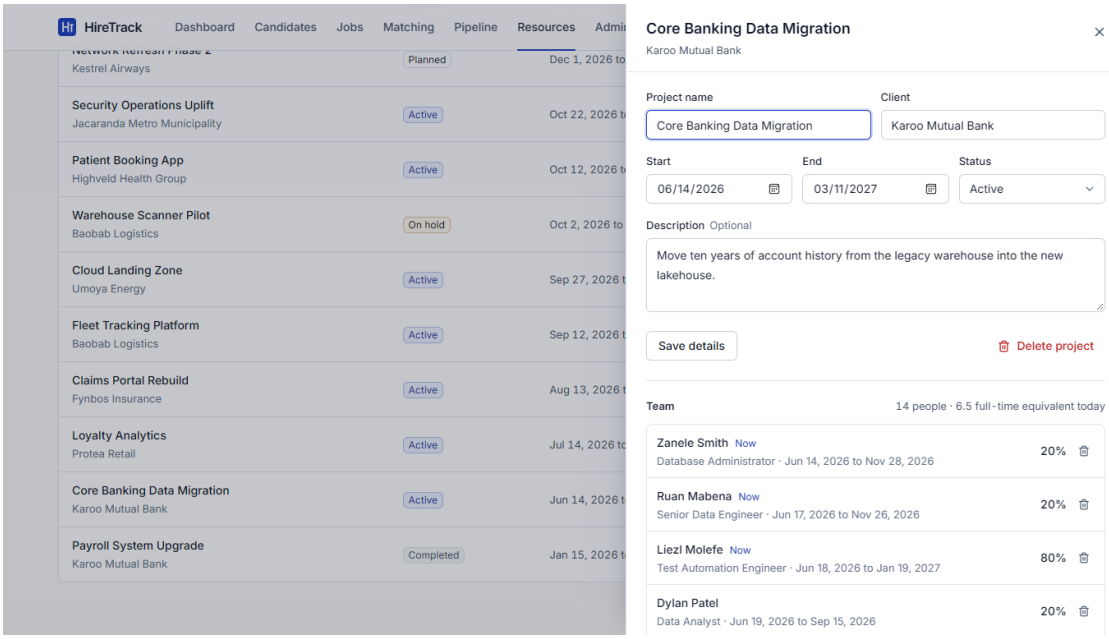


Figure 5.13: Project detail with team allocations

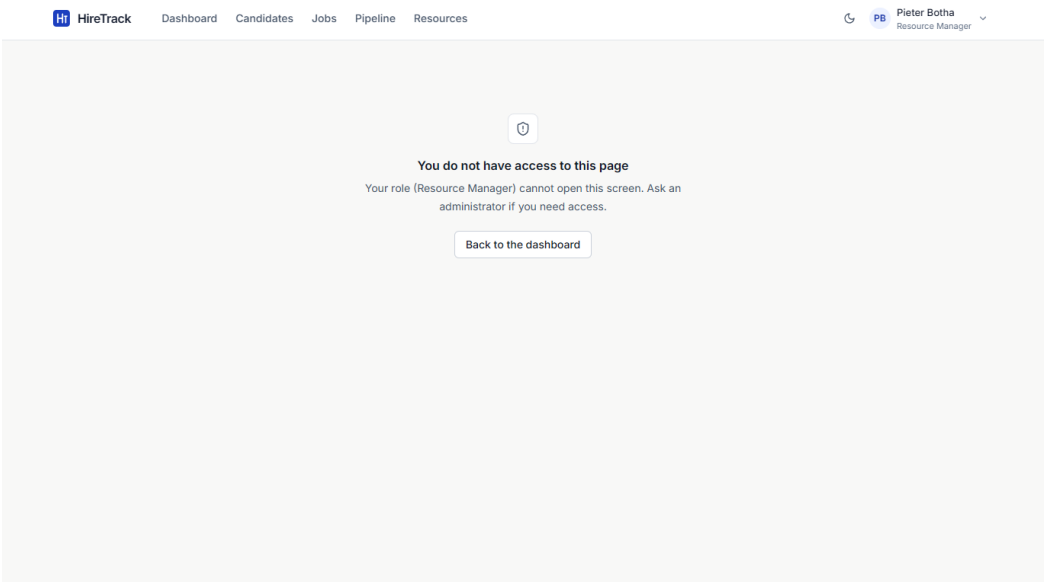


Figure 5.14: A Resource Manager opening the Matching page is refused

5.3 Testing

5.3.1 Approach

Testing followed the test pyramid: many fast unit tests of the rules, a solid layer of API integration tests, component tests of the screens, and a smaller number of end-to-end journeys in a real browser (Section 5.4). All tests are automated and repeatable.

Level	What is tested	Tool	How it is isolated
Unit	Scoring, skill normalisation, pipeline rules, allocation maths, database constraints	pytest	Pure functions; constraint tests use a temporary database
Integration	Every API endpoint, sign-in and roles, import, seed, dashboard, CLI	pytest + FastAPI TestClient	A new temporary data folder per run; every test starts from an empty database with the three accounts
Component	Sign-in, role-based menus, dashboard, pipeline, resources, import, admin, form logic	Vitest + React Testing Library	The API module is replaced with mocks
System / end-to-end	Complete user journeys (TC-01 to TC-22)	Playwright in Edge and Chrome	Own server on port 8010 with a freshly seeded database
Performance	Match speed at 50, 600 and 2,600 candidates	pytest with the real model	HIRETRACK_REAL_MODEL=1

Table 5.6: Testing levels and tools

Fast tests use a deterministic word-hashing embedder instead of the language model, so the 330 backend tests run in under a minute. Password hashing uses 4 bcrypt rounds in tests instead of 12. The performance tests switch the real model back on.

5.3.2 Unit and integration results

Test file	Level	Tests	Passed	Covers
unit/test_scoring.py	Unit	69	69	Four scores, weights, rounding, ranking and tie-breaks, aliases, explanations, fairness
unit/test_pipeline_rules.py	Unit	22	22	Stage order, allowed moves, reopen, funnel
unit/test_allocation_rules.py	Unit	21	21	Overlaps, daily load, peak load, over-allocation messages, bands
unit/test_models.py	Unit	21	21	CHECK, UNIQUE and FK rules; cascades; schema upgrade
integration/test_api.py	Integration	54	54	Candidates, jobs, matching, CSV, backups, seeding, demo
integration/test_auth.py	Integration	64	64	Sign-in, lock-out, tokens, passwords, users, role matrix
integration/test_pipeline_api.py	Integration	23	23	Applications, stage changes, hiring, shortlist from results
integration/test_resources_api.py	Integration	20	20	Staff, projects, allocations, utilisation
integration/test_import.py	Integration	20	20	CSV and Excel import, CV parsing (txt, docx, pdf)

Test file	Level	Tests	Passed	Covers
integration/test_seed_dashboard.py	Integration	9	9	Seed sizes, determinism, legal histories, dashboard
integration/test_cli.py	Integration	4	4	Password reset tool
integration/test_performance.py	Integration	3	3	Speed at 50, 600 and 2,600 candidates
Total backend		330	330	100% pass
frontend src/test (3 files)	Component	49	49	Screens with mocked API, form validation, re-ranking

Table 5.7: Unit, integration and component test results (final run)

5.3.3 Coverage

Backend line coverage is **95%** (2614 of 2748 lines), above the 70% target. The lines not covered are mainly the one-off model download script and error branches that need a missing model or a damaged disk to reach. Frontend line coverage from the component tests is 50.6%; the matching screens are exercised by the end-to-end journeys instead.

Package	Lines	Covered	Coverage
app (core: models, schemas, security, main)	889	827	93%
app/engine	422	392	93%
app/routers	240	240	100%
app/services	1197	1155	96%
Total	2748	2614	95%

Table 5.8: Backend line coverage by package (pytest-cov)

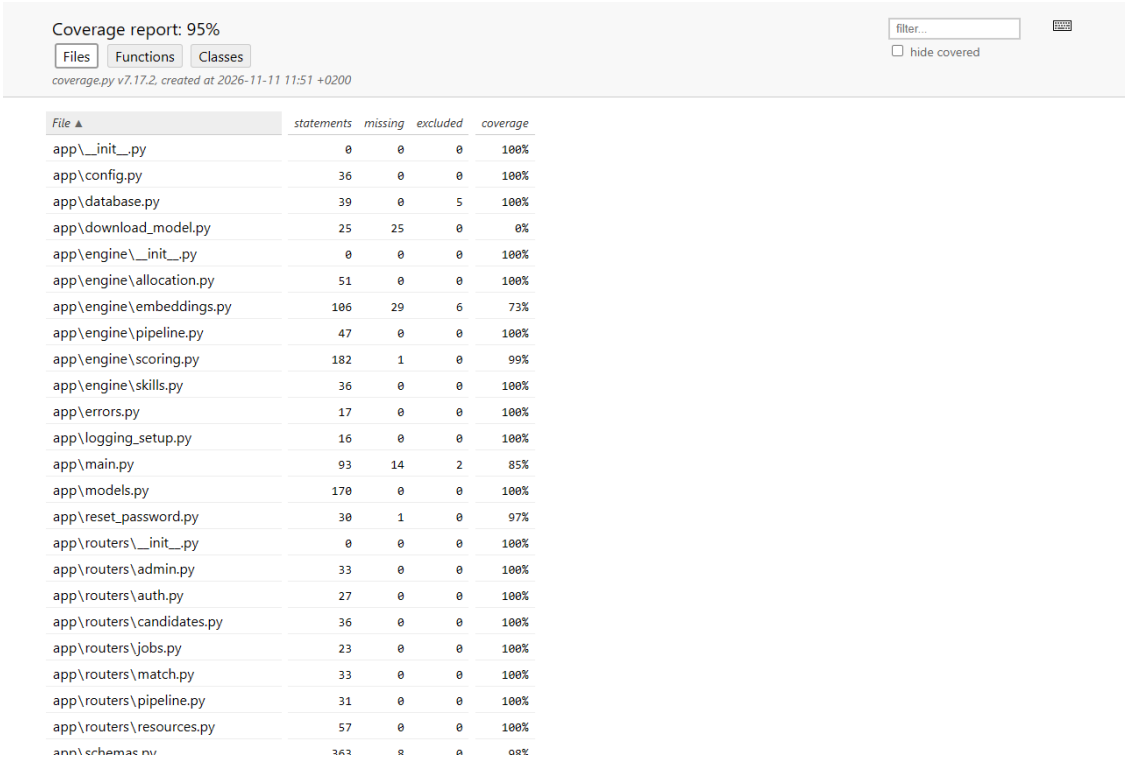


Figure 5.15: pytest-cov HTML report

5.3.4 Performance

Scenario (real all-MiniLM-L6-v2 model, 16-thread laptop CPU)	Time	Target
50 candidates, nothing cached	0.52 s	< 5 s
600 candidates, warm cache (score + save + reload)	0.16 s	< 5 s
600 candidates re-ranked through the screen (Playwright TC-13)	0.69 s	< 5 s
2,600 candidates, warm cache	0.93 to 1.06 s	< 5 s
First match after loading 600 new CVs (each CV read once)	8 to 11 s	one-off
Reading 2,600 new CVs in the background after loading	20.1 s	one-off

Table 5.9: Matching speed

5.3.5 Debugging and code improvement

Testing found seven product defects, all fixed (Section 5.4.4). Two changes improved the code beyond the fixes: the CV parser gained a vocabulary of common skills so it works on an empty database, and the seed data generator was rewritten so stage dates, hires and allocations look like a real company's history. Every fix was followed by a full re-run of all suites.

5.4 System Testing

5.4.1 Approach and requirements

System testing checks the finished system against the requirements, through the screens and the API, as a user would use it. Each test case below is traced to one or more requirements and to the automated test that runs it. The 22 Playwright journeys (TC-01 to TC-22) drive Microsoft Edge against a freshly seeded server; TC-23 to TC-36 check requirements through the API and unit suites; TC-27 and TC-37 were done by hand on a clean copy of the submission.

ID	Requirement
FR1	Add a candidate with name, contact details, location, experience, education, skills, previous titles and CV summary
FR2	View, search, edit and delete candidates
FR3	Create and edit a job specification: title, location, status, skills (required/preferred), minimum experience and education, description
FR4	Adjust the four scoring weights per job; they must total 100%
FR5-FR8	Calculate skills, experience, education and meaning (embedding similarity) scores
FR9	Combine the scores into a weighted overall score and rank every candidate
FR10	Show each candidate's score breakdown, matched and missing skills and the exact calculation
FR11	Seed or reset realistic demonstration data (500+ candidates, 30+ jobs)
FR12	Reject invalid input with a clear message and save nothing
FR13	Export ranked results to CSV
FR14	Sign in with a personal account; three roles (Administrator, Recruiter (HR), Resource Manager) with role-based access
FR15	Administrator manages user accounts; users change their own password
FR16	Capture a CV file (PDF, Word, text) to pre-fill the candidate profile
FR17	Bulk import candidates from CSV or Excel with a per-row error report
FR18	Track each application through Applied, Screened, Interview, Offer, Hired or Rejected under fixed rules
FR19	Keep an audit history of every stage change (who, when, note)
FR20	Add shortlisted candidates from match results to a job's pipeline
FR21	A hired candidate joins the staff pool
FR22	Manage staff and client projects
FR23	Allocate staff to projects by percentage and dates without booking anyone above 100%
FR24	Show utilisation per person and who is on the bench
FR25	Dashboard with key counts and charts
FR26	Back up and restore the database
NFR1	Performance: ranking 500+ candidates returns within 5 seconds
NFR2	Runs on one computer, offline after installation
NFR3	Usability: consistent screens, inline errors, keyboard and screen-reader access
NFR4	Maintainability: business rules in separate, unit-tested modules
NFR5	Privacy: only fictitious data; scoring never uses name, age or gender
NFR6	Reliability: invalid data rejected at screen, API and database level
NFR7	Transparency: every ranking explainable from visible scores

ID	Requirement
NFR8	Compatibility: current Edge and Chrome
NFR9	Security: hashed passwords, signed tokens, role checks on the server, lock-out, confirmed destructive actions
NFR10	Quality: automated test coverage of at least 70% of backend code

Table 5.10: Requirements tested (FR14 to FR26 and NFR9 to NFR10 added in this phase)

Test environment: Windows 11 laptop, 16-thread CPU, no GPU; Python 3.11.18; Node.js 20.16; Microsoft Edge and Google Chrome (current); screen 1366 x 768. Test data: the full seed dataset (600 candidates, 33 jobs, about 850 applications, 73 staff, 12 projects), plus records created by the tests themselves.

5.4.2 Test cases

ID	Req.	Description	Preconditions	Steps	Test data	Expected result	Actual result	Pass/ Fail
TC-01	FR14, NFR9	Sign-in with a wrong password is refused	App running; default accounts exist	1. Open the sign-in page 2. Enter the admin email and a wrong password 3. Select Sign in	admin@hiretrack.local / not-the-password1	Message 'Incorrect email or password'; user stays on the sign-in page	Message shown under the password field; dashboard not opened	Pass
TC-02	FR14	Each role signs in and sees only its menu items	Default accounts exist	1. Sign in as each role in turn 2. Read the main menu 3. Sign out	admin, recruiter, manager accounts	Admin: 7 items incl. Admin. Recruiter: no Admin. Resource Manager: no Matching, no Admin	Menus matched exactly for all three roles	Pass
TC-03	FR14, NFR9	Role checks are enforced by the server, not only hidden in the screen	Signed in as Recruiter	1. Open /admin by URL 2. Call GET /api/users with the recruiter's token	Recruiter token	Screen says 'You do not have access to this page'; API returns 403	Access message shown; API returned 403	Pass
TC-04	FR14	Signing out ends the session	Signed in as Resource Manager	1. Open the account menu, select Sign out 2. Open /resources by URL	-	Sign-in page shown again	Sign-in page shown	Pass
TC-05	FR1, FR12, NFR6	Invalid candidate data is refused with a message under each field	Signed in as Recruiter	1. Candidates > Add candidate 2. Leave name, education, skills empty; years = -2 3. Select Add candidate	years_experience = -2	Four inline messages; nothing saved; count stays 600	'Full name is required', 'cannot be less than 0', 'Choose an education level', 'Add at least one skill'; count 600	Pass
TC-06	FR1, FR2	Add, search, edit and delete a candidate	Signed in as Recruiter; 600 candidates	1. Add a candidate with skills Python and K8s 2. Search by name 3. Edit years to 5 4. Delete and confirm	Name 'Zinhle Test <id>', Durban, 4 years, Bachelor's	Count 601 after add; search shows 1 of 601; edit saved; count back to 600 after delete	As expected; K8s stored as kubernetes and shown as typed	Pass
TC-07	FR16	A CV file pre-fills the candidate form	Add Candidate panel open	1. Select Fill from CV 2. Choose a .txt CV	CV with name, email, phone and 'Airflow pipelines in Python and SQL on AWS'	Name, email, phone, summary filled; skills added as chips; nothing saved yet	All fields filled; chips Airflow, Python, SQL, AWS. (First run: chips were lower case - defect D1, fixed)	Pass
TC-08	FR17	Bulk CSV import keeps valid rows and reports bad rows	Signed in as Recruiter	1. Import CSV or Excel 2. Choose a file with 2 valid rows and 1 row with years = -1 3. Import file	3-row CSV	2 imported, 1 error 'row 4, Years, Cannot be less than 0'; count 602	Imported 2, errors table showed row 4; count 602	Pass
TC-09	FR3, FR4	Create a job; Run Match is locked until the weights total 100%	Signed in as Recruiter	1. Jobs > New job, fill title, location, skills, description 2. Set skills weight to 35 (total 95%) 3. Set it back to 40 4. Save	Python Developer, Cape Town, 3 years	At 95% a warning shows and Run Match is disabled; at 100% it is enabled; job saved	Warning '(now 95%)' and disabled button; enabled at 100%; saved	Pass

ID	Req.	Description	Preconditions	Steps	Test data	Expected result	Actual result	Pass/ Fail
TC-10	FR5-FR8, FR9, FR10, NFR7	Run a match and read the breakdown	Full seed data loaded	1. Open Cloud Data Engineer 2. Run Match 3. Open the top candidate	600+ candidates	All candidates ranked; breakdown shows four scores, skills and 'Overall = (...) / 100 = x'	602 ranked; top 50 shown; calculation shown and reproduces the overall score	Pass
TC-11	FR4, FR9, NFR7	What-if weights re-rank without saving	Results open	1. What if 2. Set Meaning to 60%	semantic weight 60	List re-ranks; 'Showing what-if ranking' label; saved run unchanged	Re-ranked instantly with the label shown	Pass
TC-12	FR13, FR20	Shortlist into the pipeline and export CSV	Results open	1. Tick 2 candidates not yet in the pipeline 2. Add to pipeline 3. Export CSV	2 candidates	Toast '2 candidates added'; CSV named hiretrack-cloud-data-engineer-<date>.csv with 600+ rows	As expected; CSV header and 603 lines checked	Pass
TC-13	NFR1	Re-running a 600-candidate match through the screen takes under 5 s	Model loaded, CVs cached	1. Results > Run Match again 2. Time until 'Match complete'	602 candidates	Under 5 seconds	0.68 s end to end (Edge)	Pass
TC-14	FR18, FR19, FR21	Move an applicant from Applied to Hired with notes	Candidate applied to Cloud Engineer	1. Open the card 2. For each stage add a note and select the next action	Notes for each move	Stage advances one step at a time; history shows 5 entries with notes and user; person appears in Staff as Hired	As expected	Pass
TC-15	FR18, FR19	Stages cannot be skipped; a rejection can be reopened	An application at Applied	1. POST stage=offer 2. Reject with a note 3. Reopen	-	API 422 'next stage is Screened'; rejected then back at Applied; note kept in history	As expected	Pass
TC-16	FR14, FR18	A Resource Manager can view but not change the pipeline	Signed in as Resource Manager	1. Pipeline 2. Open any card	-	History visible; no stage buttons	No action buttons shown; API also returns 403	Pass
TC-17	FR22, FR23, FR24	Create a project, allocate staff, block over-allocation	Signed in as Resource Manager	1. Add staff member 2. New project (active) 3. Allocate 70% 4. Try another 50% on the same dates	70% then 50%	First saved (0.7 FTE); second refused: 'most you can add for these dates is 30%'; staff list shows 70%	As expected	Pass
TC-18	FR22, FR12	Project end date before start date is refused	New project panel	1. Start = today+30, End = today+1 2. Create project	-	'End date cannot be before the start date'; nothing saved	As expected	Pass
TC-19	FR24	Bench filter shows only unallocated staff	Seed data loaded	1. Open /resources?band=bench	-	Every row shows 0%	All rows 0%	Pass

ID	Req.	Description	Preconditions	Steps	Test data	Expected result	Actual result	Pass/ Fail
TC-20	FR25, NFR3	Dashboard shows key counts and charts; bars are usable by keyboard	Seed data loaded	1. Sign in as Admin 2. Check tiles and charts 3. Select the Bench bar	-	4 tiles and 5 chart panels; selecting Bench opens the filtered staff list	As expected. (First run: bars were hidden from keyboard/screen readers - defect D2, fixed)	Pass
TC-21	FR15	An administrator creates a user who can then sign in	Signed in as Admin	1. Admin > Add user (Recruiter) 2. Sign out 3. Sign in as the new user	hr.<id>@hiretrack.local / Welcome123	User created; sign-in succeeds with the Recruiter menu	As expected	Pass
TC-22	FR26, FR11, NFR9	Manual backup; data reset needs the admin's password	Signed in as Admin	1. Back up now 2. Load demo set with a wrong password 3. Cancel	wrong-password1	Backup listed; 'Incorrect password'; data unchanged	As expected	Pass
TC-23	FR11	Seed script builds the realistic dataset	Empty database	1. python seed/seed_database.py	-	600 candidates, 33 jobs (30 open), 500+ applications with legal histories, staff, 12 projects, no one above 100%	600 candidates, 33 jobs, 847 applications, 1,906 events, 73 staff, 12 projects, 102 allocations; all histories legal	Pass
TC-24	FR26	Restore refuses a corrupt or unknown backup file	Admin	1. Restore a damaged file 2. Restore a path with ../	corrupt .db; '../../data/hiretrack.db'	Integrity check fails, nothing restored; traversal refused	As expected	Pass
TC-25	FR15, NFR9	There is always one active administrator; users cannot deactivate themselves	Admin	1. Change the only admin to Recruiter 2. Deactivate own account	-	Both refused with a reason	As expected	Pass
TC-26	NFR1	Ranking speed with the real model	Real all-MiniLM-L6-v2 model	Run tests/integration/test_performance.py with HIRETRACK_REAL_MODEL=1	50, 600 and 2,600 candidates	Each warm match under 5 s	50 uncached: 0.52 s; 600 warm: 0.16 s; 2,600 warm: 0.93-1.06 s; one-off encoding of 2,600 CVs 20.1 s	Pass
TC-27	NFR2	The app runs with no internet connection	Fresh install from the zip	1. Start the server with HF_HUB_OFFLINE=1 and an unreachable proxy 2. Sign in and run a match	HTTP(S)_PROXY=http://127.0.0.1:9	Everything works; no outbound connection needed	Match ran and results loaded with all network access blocked	Pass
TC-28	FR5-FR8, FR9, NFR5, NFR7	Scoring ignores personal details	-	Score two candidates identical except name and email	-	Identical scores; name used only as the final tie-break	As expected	Pass
TC-29	NFR6	Database constraints reject bad data written directly	-	Insert bad rows through the ORM, bypassing the API	years -1, weights 110%, allocation 0%, duplicate application	IntegrityError for every case	As expected (21 cases)	Pass

ID	Req.	Description	Preconditions	Steps	Test data	Expected result	Actual result	Pass/ Fail
TC-30	NFR9	Five failed sign-ins lock the account for 60 s	-	1. Five wrong passwords 2. Correct password	-	429 'Too many failed sign-in attempts'	As expected	Pass
TC-31	NFR9	Expired or forged tokens are refused	-	Call /api/auth/me with an expired and a garbage token	-	401 with 'expired' / 'sign in'	As expected	Pass
TC-32	NFR9	Passwords are stored only as bcrypt hashes	-	Read app_user.password_hash	-	Every value starts with $2b$; no plain text	As expected	Pass
TC-33	NFR6, FR12	Search text cannot inject SQL	Candidates exist	Search for x' OR '1'='1	-	No rows matched; no error	As expected	Pass
TC-34	NFR8	Full journey suite in a second browser	Chrome installed	HT_BROWSER=chrome npx playwright test	22 journeys	All pass	22 of 22 passed in Chrome	Pass
TC-35	NFR10	Backend coverage at least 70%	-	pytest --cov=app	-	>= 70%	95% line coverage	Pass
TC-36	NFR4	Business rules can be tested without a database	-	Run tests/unit (scoring, pipeline, allocation)	-	All pass with no server or database	133 unit tests passed	Pass
TC-37	Installation	Fresh install from the zip on a clean folder	Python 3.11, Node 20	Unzip, run start.bat, sign in	-	App installs, seeds and opens at http://127.0.0.1:<port>	See Installation section: completed and verified	Pass

Table 5.11: System test cases (also in system_testing/Test_Cases.xlsx with the automated test for each case)

5.4.3 Evaluation of the testing results

Pass rate. All 37 system test cases passed on the final runs: 22 of 22 end-to-end journeys in Edge and 22 of 22 in Chrome, 330 of 330 backend tests and 49 of 49 frontend tests, 401 automated tests in total. Every requirement from FR1 to FR26 and NFR1 to NFR10 is covered by at least one passing case (traceability sheet in Test_Cases.xlsx).

Not first-time results. The first end-to-end run passed 14 of 22 (64%); the first backend run passed 322 of 326. Each failure was investigated and classified as a product defect, a test defect or a test-environment issue. Raw outputs of every run are kept in testing/results.

Suite	First run	Final run
Backend (pytest)	322 / 326	330 / 330
Frontend (Vitest)	45 / 46	49 / 49
End-to-end, Edge (Playwright)	14 / 22	22 / 22
End-to-end, Chrome (Playwright)	not run	22 / 22

Table 5.12: Pass rate from first to final run

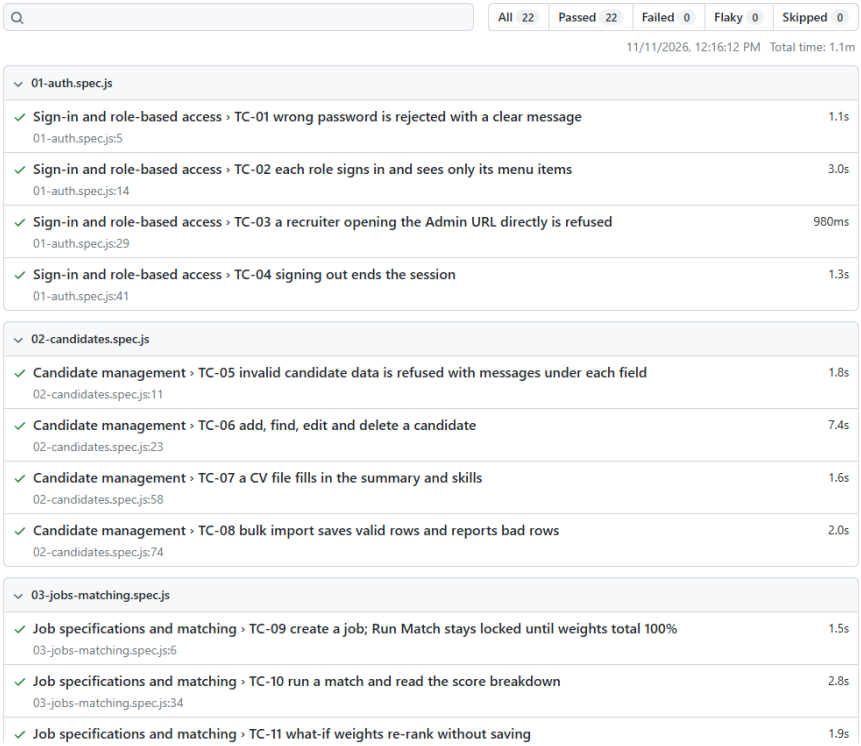


Figure 5.16: Playwright HTML report of the final end-to-end run

5.4.4 Defects found and how they were fixed

ID	Found by	Defect	Fix
D1	TC-07	Skills read from a CV appeared as lower-case chips	Converted to display labels before adding them to the form
D2	TC-20	Clickable dashboard bars were hidden from keyboard and screen-reader users	Made real buttons with names; component test added
D3	test_import	CV parsing found no skills on a new, empty database	Built-in vocabulary of common skills plus the alias table

ID	Found by	Defect	Fix
D4	Screenshot review	Seed data unrealistic: same timestamps, hires with other open offers, 59% of staff on the bench	Dates spread over time, a hire withdraws other applications, bench about 1 in 8
D5	Screenshot review	"170% booked today" on a project read as over-booking	Shown as full-time equivalents (1.7 FTE)
D6	Manual test	start.bat busy-port check never fired	Replaced the regular expression with plain findstr filters
D7	Fresh install	start.bat could not find Python 3.11 installed outside PATH	PYTHON override added to both start scripts and documented

Table 5.13: Product defects

Nine test defects were also fixed (for example ambiguous locators, an expected count that ignored records created by an earlier test, and an expectation of 29 open jobs where 30 is correct). Two test-environment issues were found. First, the end-to-end server had switched off the model warm-up the real app performs at start-up, so the first match was slow. It now starts exactly like the real app. Second, under automation Microsoft Edge holds a finished blob download open, which hung the next test. The CSV test now captures the file at the last browser step in Edge, while Chrome performs the full save.

5.4.5 Limitations

- Firefox and Safari were not tested; Edge and Chrome were.
- Usability was judged by review and accessibility checks, not by observing real recruiters.
- The first match after loading new CVs takes longer than 5 seconds because each CV is read once by the model; later matches take under a second.
- SQLite and a server bound to 127.0.0.1 suit one user at a time; concurrent load was not tested.
- Frontend component coverage is 50.6%; the older matching screens rely on end-to-end tests.
- Scanned (image-only) CVs cannot be read; there is no OCR.
- Seed data is generated, so match quality on real CVs has not been measured.

5.4.6 Recommendations

1. Add Firefox and other screen sizes to the Playwright run.
2. Hold a short usability session with two or three recruiters, using the test journeys as tasks.
3. Move to PostgreSQL and a lab-network server before several recruiters use it at once, and add a load test.
4. Add component tests for the Candidates, Jobs and Results screens.
5. Encode CVs in the background as soon as they are imported or edited, so the first match is never cold.
6. Add OCR for scanned CVs and compare match rankings with recruiter shortlists.

5.5 Installation

5.5.1 Summary

HireTrack installs on one Windows or macOS computer with a single command. The full, step-by-step guide with manual installation, test commands and troubleshooting is INSTALLATION_GUIDE.md (and .pdf) in the submission folder.

Prerequisite	Version
Python	3.11 (64-bit)
Node.js and npm	18 or later (tested with 20.16)
Browser	Microsoft Edge or Google Chrome
Disk / memory	about 3 GB free; 4 GB RAM minimum, 8 GB recommended
Internet	During installation only

Table 5.14: Prerequisites

5.5.2 Steps

1. Unzip `402311822_HireTrack_Phase5_Implementation.zip` and open `source_code`.
2. Windows: run `start.bat`. macOS/Linux: run `chmod +x start.sh` then `./start.sh`.
3. The first run creates the Python virtual environment and installs the packages, downloads the language model (about 90 MB), builds the screens, and creates and seeds the database. Later runs start in a few seconds and work offline.
4. Open `http://127.0.0.1:8000` (the script opens it) and sign in with one of the accounts below.

Role	Email	Password
Administrator	admin@hiretrack.local	Admin@123
Recruiter (HR)	recruiter@hiretrack.local	Recruiter@123
Resource Manager	manager@hiretrack.local	Manager@123

Table 5.15: Default accounts (change the passwords before real use)

Options: `PORT` (another port), `PYTHON` (path to Python 3.11 if not on PATH) and `HT_NO_BROWSER`. The database can be re-seeded at any time with `python seed/seed_database.py` or from the Admin screen.

5.5.3 Verification on a clean copy

The guide was verified by doing a fresh installation from the submission contents in a new temporary folder on the test laptop, with no virtual environment, no node_modules, no model and no database. start.bat was run on port 8002. It installed the packages, downloaded the model, built the frontend, seeded 600 candidates and 33 jobs, and started the server. All three accounts signed in, a match ran, and the dashboard, pipeline and resources screens loaded. The server was then restarted with HF_HUB_OFFLINE=1 and all HTTP and HTTPS traffic sent to an unreachable proxy. Matching and every screen still worked, which confirms NFR2 (offline after installation). The install log is in testing/results/fresh-install-log.txt.

Conclusion

The Implementation Phase delivered a complete, working HireTrack. The matching engine from the Design phase is built exactly to its specification and explains every score. Around it, the system now has the remaining objectives of the Proposal: personal accounts with three roles, an applicant pipeline with a full audit history, resource allocation that cannot over-book anyone, bulk import and CV capture, and a dashboard. The database seeds itself with 600 candidates, 33 jobs and a realistic hiring and project history.

The system was tested at four levels with 401 automated tests and 37 traced system test cases. Every requirement passed on the final runs, with 95% backend coverage, and a 600-candidate match takes well under one second once CVs are cached. Testing found and fixed seven product defects; the remaining limitations (browser range, cold first match, single-user database) are documented with recommendations. A fresh installation from the submitted files was verified with one command.

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